

Title: Biosystems modeling of cell-cell communication (BioComm)

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

<i>Name*</i>	<i>Institution</i>	<i>Department</i>	<i>Email</i>	<i>Tenured / Tenure-track</i>	<i>For Erasmus MC applicants: clinician/ researcher/ mix</i>	<i>Role in Flagship</i>
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Corresponding Lead: Kerstin Wendt

Planned start date of seed funding: 1-12-2022

Primary Convergence theme: *Fundamentals of Health & Disease*

Secondary Convergence theme (*if applicable*): Choose an item.

Tertiary Convergence theme (*if applicable*): Choose an item.

Quaternary Convergence theme (*if applicable*): Choose an item.

Click or tap here to enter text. **Keywords:** *Microengineering, Organoids, 3D cell microenvironments, Mathematical modelling, Disease modelling, Artificial membranes, Organoids & Philosophy, Drug screening*

1.b Summary (max 300 words, now 299)

The medical field advances rapidly, and patients with complicated medical problems have increasingly higher chances of survival, leading to new challenges. In addition, every patient is unique in her/his response to treatment modalities, increasing the emphasis on personalized treatment. These challenges require an increment in our fundamental knowledge of organ development and homeostasis in health and disease.

Recent advances in tissue/organoid technology open up unprecedented possibilities to study the fundamental principles of developmental biology and how interference with these principles leads to disease. However, heterogeneity in current cultures and incomplete modeling of the in vivo physiology still hampers broad implementation of organoid technology in the clinical setting.

Our interdisciplinary BioComm flagship plans to solve these hurdles by convergence of state-of-the-art expertise at three institutes, integrating developmental and cell biology, genomics, biophysics, micro-engineering, mathematical modelling, epidemiology, psychology, philosophy and ethics. BioComm aims to develop multi-cell lineage, complex biosystems to increase our understanding of the fundamental self-organizing principles of brain and lungs during development and in tissue homeostasis. We will unravel how perturbation of these principles leads to disease, and use this knowledge to improve diagnostics and therapeutics.

The first steps towards a long-term interdisciplinary research program will be taken in five interconnected work packages covering different themes; including micro-engineering and organoid technology, multi-dimensional data integration and modelling, and artificial membrane technology. The developed biosystems will subsequently be challenged by genetic disease variants and prenatal exposures, used for drug discovery applications, and in knowledge utilization and translation using a societal and ethical perspective.

BioComm's ambition is to establish a long-term interdisciplinary research and training initiative, anchored into existing programs, for talented (under-)graduate students and researchers. The consortium will dedicate its outreach and dissemination activities to stakeholders including fundamental biomedical researchers, bioengineers, computational scientists and social scientists - within academia and industry context.

1.c Public Summary (max 300 words, now 257 words)

Patient care is increasingly focused on treating severe and complicated patients, as well as individualized care, so called personalized medicine. Therefore, it is imperative to improve our current understanding of development and homeostasis of organs in health and disease.

The ambition of this flagship program is to establish a sustainable interdisciplinary research network that aims to use fundamental research to innovate and improve medical care.

We are developing improved, sophisticated miniature versions of human organs in culture, based on organoid technology, which will be then used to model diseases and the influence of stress like drugs. These so-called biosystems of the brain or the lungs can inform us on fundamental processes in the human body, increase our understanding of how these organs develop and function normally, in healthy conditions, but also of how certain diseases (e.g., brain tumors, epilepsy, or neurodevelopmental diseases, or asthma) develop. We will use our data to build computer models for how these organs develop normal and predict responses to disease factors, with the aim to develop new treatments.

Team science with excellent researchers from clinics, cell biology, genetics, engineering and computation is central to tackle these fundamental research projects and develop also a strong teaching program aligning to our research with courses and student projects in bachelor and master programs at the Erasmus MC, TU Delft and Erasmus University Rotterdam.

Translation of knowledge on fundamental processes to clinical practice and acceptance of new technologies by the general audience can be challenging, and therefore we will also study philosophical, ethical, and societal aspects.

2. Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

Recent advances in organoid technology have rekindled interest in the basic principles of developmental biology and how interference with these principles leads to disease. However, broad implementation of organoids, for example in personalized drug screening platforms, is hampered by batch-to-batch variability and incomplete modeling of the *in vivo* physiology.

Our BioComm flagship aims to overcome these hurdles by developing next-generation human tissue models with the help of microengineering ('biosystems') (Fig. 1) with standardized readouts and mathematical models to accurately predict tissue development. We will use these biosystems to delineate novel principles of cell-cell communication important for human tissue development and model disease-linked tissue changes caused by genetic conditions or triggers (e.g., hypoxia, drugs) that were identified in population studies as relevant for brain and lung development.

Achieving this requires the combined expertise of the EMC (organoid culture, disease modeling), TU Delft (microengineering, mathematical modeling) and the EUR (population studies, societal aspects of biosystems) (Fig. 2).

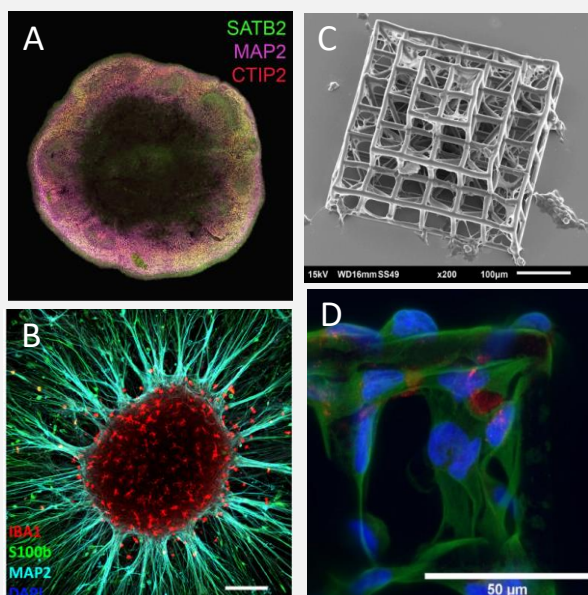


Figure 1: Organoids and scaffolds

A) Brain organoid (70 days) with necrotic core stained for neuronal markers (**DvdB**).

B) Adherent cortical organoids showing neuronal dendrites (light blue), astrocytes (green) and microglia (red) (**FdV**).

C) SEM micrograph of a polymeric micro-scaffold onto which primary patient derived glioma cells are seeded and cultured (**AA**, **PF**). D) Example of patient-derived cells grown onto a micro-scaffold and stained for EGFR (red), dapi (blue) and tubulin (green) (**AA**, **PF**).

With these ambitions our flagship contributes to all four Health&Technology missions:

- *Life-long health, from pre-birth to end of life:* early life insults may have profound health effects later in life. BioComm focuses on brain and lung, and aims to develop advanced biosystems to better understand cellular processes involved in (ab-)normal development. Insights into these mechanisms will also contribute to understand, model and explore homeostasis and aging.
- *Socio-economic equality in health(care):* although our flagship is a fundamental research consortium, the development of advanced brain and lung biosystems may facilitate the introduction of personalized medicine in the clinic. As such, BioComm may raise awareness and investigate how personalized medicine may affect the socio-economic gap in healthcare with our research component around expectation management.
- *Prevention and early diagnostics to proactively maintain health:* early disease detection will assist in improved healthcare, as late diagnosis affects treatment options. Our biosystems will be used to study early effects on abnormal growth, and may therefore contribute to improved diagnosis at an early stage, by identifying relevant stage-specific genes/mechanisms and biomarkers/molecules. As example, for neurodevelopmental syndromes the genotype-phenotype links are often unclear. Insight into disease mechanisms can provide parents with a clear prognosis for their child's development and determine treatment options. Moreover, lung biosystems will be used to identify genetic/environmental risk factors and may in the future also help to identify pediatric lung patients at risk of developing chronic disease or comorbidities, leading to improved monitoring and care.
- *Individual-tailored precise medical treatment for all:* our flagship aims to develop brain and lung models in which patient-derived cells may be directly tested for responsiveness to specific treatment modalities. This could result in tailor-made therapeutic approaches for individual patients or patient-groups.

BioComm naturally fits in the 'Fundamentals of Health & Disease' theme, as it aims to increase our understanding of early human tissue development and disease. The next-generation biosystems that BioComm will generate, will provide important tools needed to realize various aims of Convergence H&T, in particular the need for better model systems to enable early diagnosis and personalized treatments.

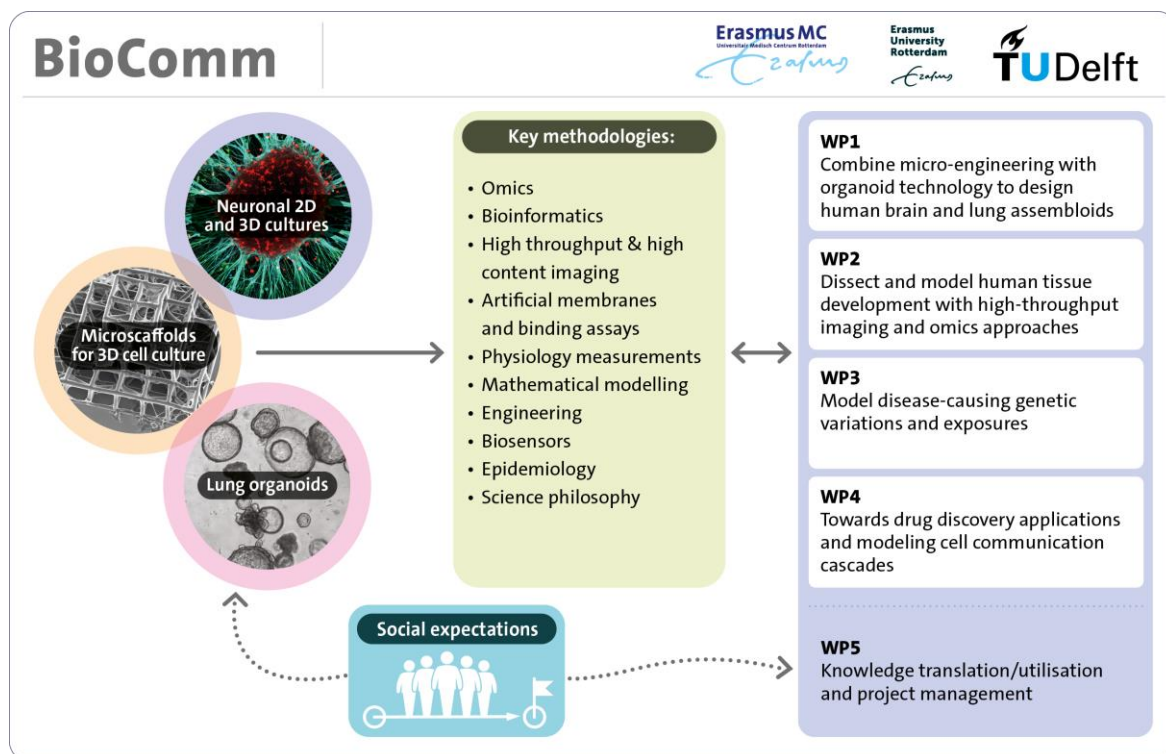


Figure 2: Integration of biosystems, state of the art analysis tools and societal considerations in BioComm

2.b Synergy between disciplines

BioComm integrates expertise on organoid culture, engineered micro-structures and biosensors to improve standardization and create biosystems with increased complexity in a reproducible manner. Mathematical modelling on acquired multi-dimensional data (e.g., -omics, imaging) will allow us to define and refine principles of human physiology and disease pathogenesis. Key to achieving these ambitious goals is the combination of state-of-the-art knowledge in cell and molecular biology, micro-scale engineering, computational science, and mathematics. To understand and make use of the societal potential and consequences of our ambitious program we incorporated sociological, philosophical and ethical expertise. BioComm provides a framework for connecting and efficiently integrating all these disciplines, which are contributed by excellent research groups at the participating institutes.

Key examples of synergy in BioComm:

- *Building next-generation 'biosystems'.* Developmental biologists (EMC) and micro-engineers (TUD) combine biological (e.g., cell isolation, growth factor selection, organoids, iPSC) and engineering (e.g., cell instructive scaffolds, specific coatings, biosensors) expertise to develop advanced biosystems from existing brain and lung organoid cultures established at the EMC.
- *Biological data interpretation, extrapolation and modelling.* Cell and molecular biologists (EMC) will acquire large amounts of data to describe cellular behaviour. These data need to be processed,

interpreted and compared using the unique and specialized expertise of computational, imaging, and mathematical experts (EMC/TUD). Full comprehension of integrated datasets will result in brain and lung models that can be used to predict the outcome of exogeneous stimuli on individual systems.

- *Use biosystems to understand the impact of early-life exposures on organ development.* Biomedical scientists (EMC), epidemiologists (EMC) and clinical psychologists (EUR) will address key clinical/societal issues. For example, brain biosystems will be exposed to hazardous substances that were identified in the large population-based Generation R study as potentially detrimental for brain development and function. Lung biosystems will be generated with patient-derived primary or iPS cells, or exposed to hypo/hyper-oxic conditions to mimic pediatric lung diseases. The biosystem's responses will be monitored with microsensors (TUD) and molecular analyses (EMC). Findings from these studies could in return be used to draft new questions for implementation in Generation R, creating unique synergy between two disciplines.
- *Personalized medicine, drug and biomarkers discovery:* Using patient derived cells, the brain and lung biosystems are ideal advanced models to study patients' responses towards specific therapeutic agents (EMC/TUD). Biosystems can also be used to perform large drug or toxicity screens in collaboration with pharmaceutical industry and other flagship initiatives working towards such assays. Finally, biosystems can be manipulated to mimic diseases and as such be used to embark into biomarker discovery (EMC/TUD/EUR).

2.c Long term sustainability plan

Program Feasibility

Our flagship combines state-of-the-art technology and biology to develop advanced brain and lung biosystems that will be at the forefront of science.

While ambitious in nature, feasibility of BioComm is safeguarded by:

1. The expertise, experience and commitment of the individual PIs and their institutes. Importantly, this includes involvement of other networks built by individual PIs in the H&T Convergence program. Together, this will ensure a state-of-the-art and internationally competitive research framework.
2. All necessary infrastructure is available at the partner sites and new state-of-the-art equipment will become available, for example by embedding of the Erasmus MC proteomics Center (**JD**) in the NWO LRI roadmap.
3. Existing small-scale collaborations of BioComm PIs in both research and education have already shown excellent collaborative potential.
4. All partners are experienced in guiding BSc/MSc/PhD-students and are involved in different educational programs.

Long-term sustainability

Collaboration between individual research groups warrants the emergence of new and exciting research lines; envisioned examples include i) the use of brain/lung biosystems to develop new high-throughput drug screening platforms (EMC/TUD), ii) new knowledge and tools to inform future therapy designs e.g. (personalized) prevention strategies and diagnostic methods (EMC/TUD), iii) the use of acquired expertise with 3D biomaterials to explore new research lines linked to implant-based neural tissue engineering (e.g. for stroke, spinal cord injury), and iv) the interplay between biosystems and large population-based cohort studies (e.g. Generation R) to open possibilities for combined biomedical/epidemiological studies into other health hazards (EUR/EMC).

New long-term collaborations between consortium members, with fellow flagship initiatives and beyond will arise that will provide ample opportunities to file joint applications for external funding beyond the first 5-year period; examples are curiosity-driven calls (e.g. NWO ENW-XL consortium grant), disease-focused calls (e.g., Lung Foundation Netherlands consortium grant, Horizon) and technology-focused calls at NL/EU levels (e.g. NWO Roadmap Large-scale infrastructure).

Moreover, we foresee that this flagship will provide talented early-career researchers with an attractive and competitive skillset that will foster successful applications for personal grants (e.g., now-Veni/Vidi, ERC Starting Grant).

By anchoring BioComm into existing teaching programs and by developing new educational modules to attract talented researchers we will create impact on educational programs of EMC/EUR/TUD, effectively integrating our convergence program into the educational system (also see below).

2.d Visibility, attracting talents and new partners

Visibility

BioComm's goal of building a next generation of human biosystems has enormous potential for basic and clinical sciences - and eventually healthcare and society (section 2e). The power of this advanced combination of cell biology and technology will capture the imagination and interest of the general public, and of young and talented researchers. In WP5 we will carry out a dedicated program for teaching, dissemination and outreach to enhance awareness of our biosystems and their possible impact but also relate to the societal impact of our research. We will reach the general via social media channels and dedicated outreach platforms such as "Amazing Erasmus MC" and "TU Delft stories".

For students, we will create teaching modules and incorporate these into existing BSc/MSc programs, enhancing the visibility of the program to those groups.

Engagement of students and early career researchers

We will attract BSc/MSc students to work on small scale projects in the program (e.g., modelling, imaging) lasting 3-9 months. These students will benefit tremendously from the dynamic cross-disciplinary research team we have built in which biomedical, technological and social competences act in synergy.

BioComm PIs are longstanding active contributors in BSc (e.g., Nanobiology, Psychology) and MSc programs (e.g., Molecular Medicine, Nanobiology, Life Science and Technology, Clinical Psychology, Philosophy). **TI** is head of the MSc Nanobiology program.

Attracting talents

From experience, the majority of student interns strive to do a PhD. As we have internship opportunities at all three institutes, we will be able to offer PhD positions to the most talented students.

Most importantly, we will apply for a Marie Skłodowska-Curie Cofund Action for PhD students, which provides funding for regional programs for training and career development, through co-funding mechanisms. Chances for success are very good since the environment between EMC, EUR and TU Delft with the ongoing graduate programs and ongoing collaborative programs are ideal. Two application rounds (2023 and 2024, application in spring, decision in August) are possible within the Convergence period of 5 years. This program will allow us to recruit a large number of international PhD students. Since one MSCA Cofund application can receive up to €10 million Euro we aim to include other flagship programs that align with our program in this initiative. The Rare

Disease Innovation Center (RDIC) initiative confirmed already interest. BioComm plans to establish at least 10 additional PhD positions.

Attracting partners and funding

Initially, the networks of individual PIs will be used to attract and involve clinical associates, companies and patient associations. Some are already collaborating with clinicians (**RR, RS, SB, KW, TVH**), companies (**RR, DMM**) and patient associations (**RR, SB, KW**), but the aim of the flagship is to extend these individual collaborations towards an embedded partnership.

Teaming up with other flagship programs for the MSC Cofund Action will further strengthen the Convergence and open up more possibilities for future collaborations and joined funding applications (section 2c).

2.e Long term societal impact

BioComm will provide the unique cross-disciplinary scientific environment required to construct a next generation of model biosystems for studying human organ development and disease. Direct scientific impact encompasses unprecedented fundamental insight into how human brain and lung development is orchestrated by cell-cell communication and how these processes are perturbed in cancer, congenital defects or due to early life exposures. Importantly, our biosystems and mathematical models will empower other scientists to construct their own next generation model systems for studying human organ (patho)physiology. Beyond gathering new insights into fundamental principles underlying human life and disease, BioComm will pave the way for the generation of *in vitro* avatars of specific diseases or even individual patients, allowing for: i) new opportunities to investigate and diagnose rare diseases, ii) develop biomarkers or explore new therapeutic angles, iii) predict therapy response or the detrimental impact of early-life exposures, and iv) for conducting large scale drugs screens. Hence, BioComm has the potential to prevent unnecessary interventions to reduce side-effects and health care costs. Moreover, our biosystems will reduce the need for animal model systems. Stakeholders thus include fundamental biomedical researchers, bioengineers, clinicians, computational and social scientists - both within academia but also in an industry context. Currently, most of the technology is at the TRL 2-4 level, indicating the need for more development in designing 3D culture biosystems before it may be implemented as a medical culture and testing system. BioComm aims to propel the biosystems towards the use as disease models for personalized medicine, which is TLR5-7. However, some technological advances (e.g., particular cell types/scaffold combinations, biosensors) may have broader applicability and as such will reach higher TLR levels, including patenting and commercialization.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

The **major scientific ambition** of the BioComm flagship is to generate a better understanding of the fundamental self-organizing principles of brain and lung during development and in tissue homeostasis. Particular emphasis will be placed on cell-cell communication, an essential aspect of self-organization that is regulated by signaling pathways, which often contain suitable points for pharmaceutical intervention. Brain and lung were selected as initial focus because of our proven expertise with these systems and because disease-related disturbances in these tissues have a strong impact on the life of the affected individuals as well as on society. The **clinical ambition** of the BioComm flagship is to unravel how perturbation of self-organizing principles in brain and lung leads to disease, and use this knowledge to develop improved diagnostics and therapeutic options. A **long-term ambition** of the BioComm flagship is to develop model systems of brain and lung that accurately mimic the *in vivo* situation. These have the potential to be adapted (in collaboration with other groups and/or flagship initiatives) into a personalized drug screening platform that can be used to test and predict clinical outcome of patients, leading to improved and personalized medical care. Finally, our biosystems can be used to model and experimentally validate the outcome of specific environmental insults (e. g. (viral) infection, medicinal and recreational drug exposure, nutrition imbalance, air pollution). We will integrate knowledge from our model systems with multi-dimensional population-based data to guide focused prevention strategies that aim to reduce social inequality in health.

3.b Aims and feasibility

The **overall aim** of the BioComm flagship is to engineer a new generation of human brain and lung 3D models, defined here as '**biosystems**', and study fundamental principles underlying brain and lung self-organization and the consequences of genetic or environmental perturbations.

Specific objectives are:

1. Build optimized brain and lung biosystems through integration of state-of-the-art cell and organoid culture with microengineering (sensors, scaffolds) for 3D culture.
2. Address cell behaviour and communication during tissue self-organisation from the molecular scale to that of the entire system and develop mathematical models.
3. Challenge our biosystems with pathogenic gene variants or environmental perturbations and link outcomes to patient and population data.
4. Assess the philosophical, ethical and societal aspects of organoid technology.
5. Integrate BSc and MSc students into our program with projects and teaching activities (detailed in 2D and implemented by WP5).

Feasibility:

As described in more detail in section 4b, the expertise required to challenge the aims of the flagship are already present in the participating research groups, both from the biology as the technology perspective.

Staff and budget:

BioComm will start in year 1 with 4 Postdocs (2*0.5 FTE and 2*1 FTE, one established with Convergence funding), 3 PhD students and 3 technicians (1*0.2 FTE and 2*1FTE via Convergence funding). Altogether, the BioComm partners have committed 1.278 k€ in kind contributions to the project.

To establish and, importantly, maintain the improved organoid cultures over the entire project we will invest part of the Convergence contribution to create positions for well-trained technicians. The

engineering of scaffolds requires specific expertise and experience, we will therefore appoint postdoctoral researchers here. Further we will invest in a bioprinter to create scaffolds and cover genomic sequencing costs for the characterization of the biosystems developed.

From year 2 on, we will expand our program by creating positions with additional external funding, planned is an MSCA Cofund application among other initiatives (outlined in 2d).

Possible risks:

The expertise of the BioComm partners covers all aspects of the intended projects. Although we consider the risk low, the only major risk is that the establishment of the intended biosystems in WP1 might take longer than intended since we need to explore a lot of possibilities offered by microengineering and the differentiation of iPSC's into neuronal structures is time-consuming. Therefore, we are planning to start the projects planned for WP2 and WP3 with already established culture systems, e.g., the adherent cortical organoid model (Fig. 1B) and existing data, e.g., for mathematical modelling.

3.c Work plan

Our ambitious objectives will be pursued in dedicated work packages. See also Fig 5 for the collaborations between the Partners and institutes.

WP1: Combine micro-engineering with organoid technology to obtain human brain and lung assembloids ("biosystems") (heads: AA, DvdB, RR).

We will develop simple brain and lung 3D engineered cell-culture models (Fig.1C-D) followed by the generation of complex brain and lung biosystems. Here we combine our expertise in cell/organoid culture with micro-engineering of multi-scale 3D scaffolds as well as culture surfaces to guide the assembly of tissue-mimetic structures for human brain and lung.

Scaffolds with different shapes (microns-sized to millimeter sized) and porosities will be generated (two-photon polymerization, stereolithography) (**AA**) and coated with various types of extracellular matrix or "membrane-like" coatings containing cell surface receptors (**DMM**). Primary or induced pluripotent stem cells (iPSCs)-derived organ-specific cells will be added to those scaffolds and differentiated³⁻⁵ (**RR, DvdB, FdV, SB, PF**). To continuously monitor culture parameters such as pH, oxygenation level and temperature we will integrate biosensors in the culture vessels⁶.

Subsequently we will increase tissue complexity by including specialized cell types (e.g., microglia and other types of immune cells, pericytes and endothelial cells) and, as future perspective, by fusing different organoids (e.g., blood vessel with brain or lung) to obtain complex assembloids.

We will use these engineered biosystems (organoids and assembloids) to study normal organ development (WP2) and to model disease/environmental perturbations (WP3).

WP2: Dissect and model human tissue development with high-throughput imaging and omics approaches. (heads: TI, FdV, RS)

At the start, we will use existing data from brain and lung organoid cultures (e.g., scRNA-Seq, imaging data), either from our own labs or published data from similar projects⁷⁻¹², to build the initial mathematical models (**TI, SK**).

We will apply and expand these models during the project with data obtained from our biosystems (see also WP3). For example, fluorescently labelled iPSCs carrying a genetic variation affecting neuronal development and control iPSCs will be combined at different ratios to generate sparsely labelled organoids in which cell morphology, migration, positioning and interactions can be analysed in detail over prolonged periods of time using advanced imaging techniques (Opera Phenix system or light sheet microscopy)¹³ (**FdV, DvdB, RR**) Functional analysis of will be performed by electrophysiology measurements for brain (whole-cell-patch-clamping, fluorescence-based calcium imaging and multi-electrode arrays), and TEER (Trans epithelial electrical resistance) measurements as well as oxygen sensors for lung organoids. Additional information on spatial distribution and cell identity will be obtained from spatial transcriptomics (**WvIJ**) and imaging mass cytometry (**RS**).

Active signaling pathways and morphogen gradients identified by single-cell proteomics (**JD**) will be integrated into the models. Physiological parameters, such as pH or oxygen levels, will be monitored in real time using integrated sensors. We will subsequently use the models to derive crucial parameters determining mutant cell behavior and predict outcomes of environmental perturbations, which will be verified experimentally and in population-based cohort data.

WP3: Model disease-causing genetic variations and exposures (heads: **RS, SB**)

We will model genetic variants impacting neuronal development (e. g. microcephaly/macrocephaly and complex syndromes, for example Cornelia de Lange Syndrome (**DvdB, KW**). Importantly, our biosystems will enable us to model somatic mosaicism, a frequent feature of monogenic diseases of the central nervous system but the impact for neuronal development is not understood at all¹⁴. Healthy and mutant cells (fluorescently labeled) will be combined to grow mixed brain organoids (Fig. 3) to observe interactions between mutant and healthy cells and their contribution to different tissues (live cell imaging, single cell RNA-seq, epigenome, transcription factor binding, non-cell autonomous functions) (**SB, DvdB, KW, FdV, WvIJ, HvdW, JD**). For comparison with in vivo brain development a similar approach will be used in chimeric/mosaic zebrafish embryo's (**TvH**).

Exposure of lung biosystems to varying oxygen levels, specific allergens and/or cytokines will be used to mimic pathological lung remodeling as observed in bronchopulmonary dysplasia or asthma, allowing us to compare epithelial cell behavior, endothelial-epithelial crosstalk and epithelial-immune cell crosstalk in healthy versus disease contexts (**RR, RS**).

Further, we will model cellular interactions in glioma, the most common malignant type of primary brain tumors. The 3D-engineered culture models for gliomas will accurately mimic the tri-dimensional spatial configuration followed by cells in-situ, allowing to dissect cell-cell communication (e.g., through so-called tunneling nanotubes and tumor micro-tubes) and signaling (**AA, PF**).

Large population-based studies (Generation R) showed that prenatal exposure to tobacco and cannabis alters prefrontal cortex development¹⁵. We will mimic such prenatal exposures with our brain assembloids to decipher the molecular pathways underlying these changes in parallel to *in-vitro* studies in WP2 (**HEM, IF, DvdB, FdV**).

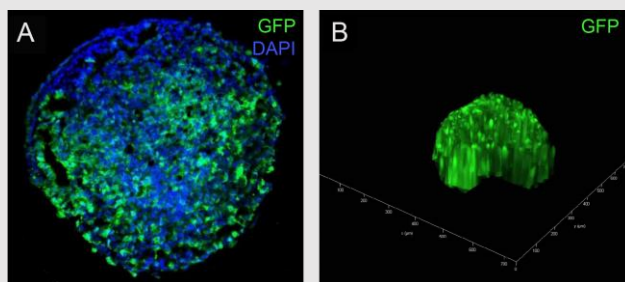


Figure 3: Mosaic brain organoids grown from a mixture of fluorescent-labelled (green) and unlabelled iPSC's. A) Confocal microscopy image and B) light sheet microscopy image (**DvdB**).

WP4: Towards drug discovery applications and modeling cell communication cascades (heads: **DMM, RR, DvdB**)

Parallel to the studies on biosystems we will create biomimetic model membrane systems for drug testing against membrane-bound receptors that are found in lung and brain. A novel rapid membrane protein reconstitution methodology (patent pending, **DMM**)¹⁶ allows to create biological membranes embedded with relevant surface receptors in a microfluidics device.

We will embed cannabinoid receptors (CB1 and CB2) and screen the cannabinoid library provided by Biopharm HK Ltd to relate binding strength to a receptor to addiction phenotypes in the brain (**DMM, HEM, IF**). This is tightly linked to mimicking prenatal exposures in WP3.

A similar approach will be applied to important receptors, such as PDGFR α , in lung development (**RR**, **DMM**). Moreover, the lung biosystems will be employed to screen orphan drugs for use in pediatric lung disease (**RR**).

WP5: Knowledge translation/utilization and project management (heads: HEM, NG, KW)

Organoids as well as our biosystems pose important questions from a science philosophical perspective. Awareness of scientists to the philosophical, ethical and societal aspects is important to effectively integrate and respond to societal expectations, views and concerns.

For the PhD students engaged in BioComm we plan a 2-3 months project to explore the ethical and science philosophical angle of their project and co-author a publication on this topic (**HZ**). Out of this, an ethics module for training sessions/meetings of the consortium will be developed.

Further, BioComm will develop education material and teaching programs for bachelor, master and PhD students in the medical, psychology and biotechnology curricula from the EUR, EMC and TU Delft (**all partners**).

BioComm will engage the public for example by organizing science classes on primary schools. These will be developed together with the “Wetenskapsknooppunt” at the EUR (**HEM**). To professionalize these science classes and dissemination materials, we will apply for grants supporting science communication projects.

This WP will actively support the dissemination activities of all partners (social media, activities in Museums, Brain awareness events etc.) by a dissemination plan and a BioComm webpage linked to webpages of other Dutch and international initiatives. Also, contributions for the outreach platforms “Amazing Erasmus MC” and “TU Delft stories” will be provided.

This WP will also manage organizational aspects and relates to the overall management of the flagship (**KW, HEM, DMM, NG, JG**) (see 4d) and coordinate the MSCA Cofund application as well as the other funding applications.

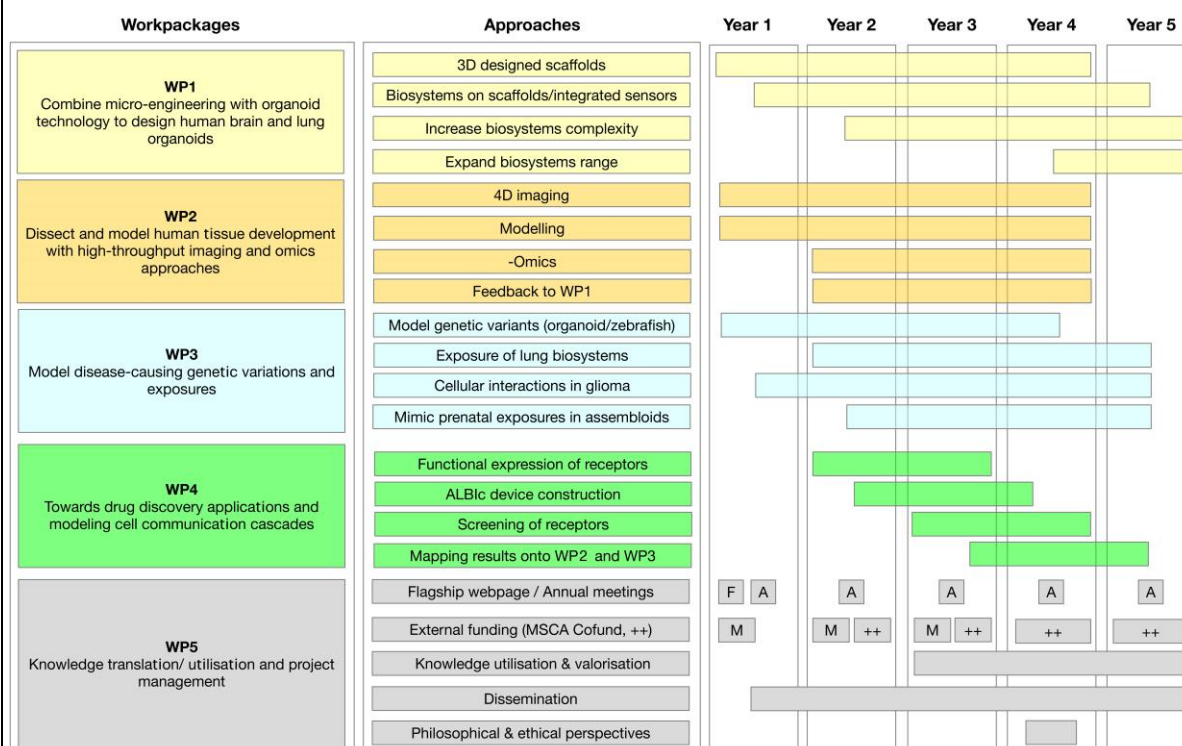


Figure 4: Timeline of the activities in the different work packages.

3.d Alignment of the program with relevant national and international research agendas

Generating optimized biosystems is pursued by many labs worldwide, as it holds tremendous promise and has far reaching implications for population health and personalized medicine. BioComm aligns in the Netherlands with the Gravity program “Organs-on-chips” (awarded 2017) headed by Christine Mummery (LUMC) and including co-applicants JG and FdV. Benefiting from the output of this program, our flagship takes the next steps by combining different cell lineages into complex biosystems and by describing tissue development with mathematical models.

The objectives of our program (see 3b) are in line with the NWA (Dutch Science Agenda), for example questions (#101) on modelling the human body for health research, (#082,#083) on development of the nervous system and prevention/treatment of neurological and psychiatric conditions, (#089) on advancing insight and treatment of pulmonary diseases, (#095) on the implementation of personalized medicine.

At the European level, our flagship aligns well with the longstanding EU research focus on rare diseases, now catalysed by a European Partnership and Joint Programme on Rare Diseases.

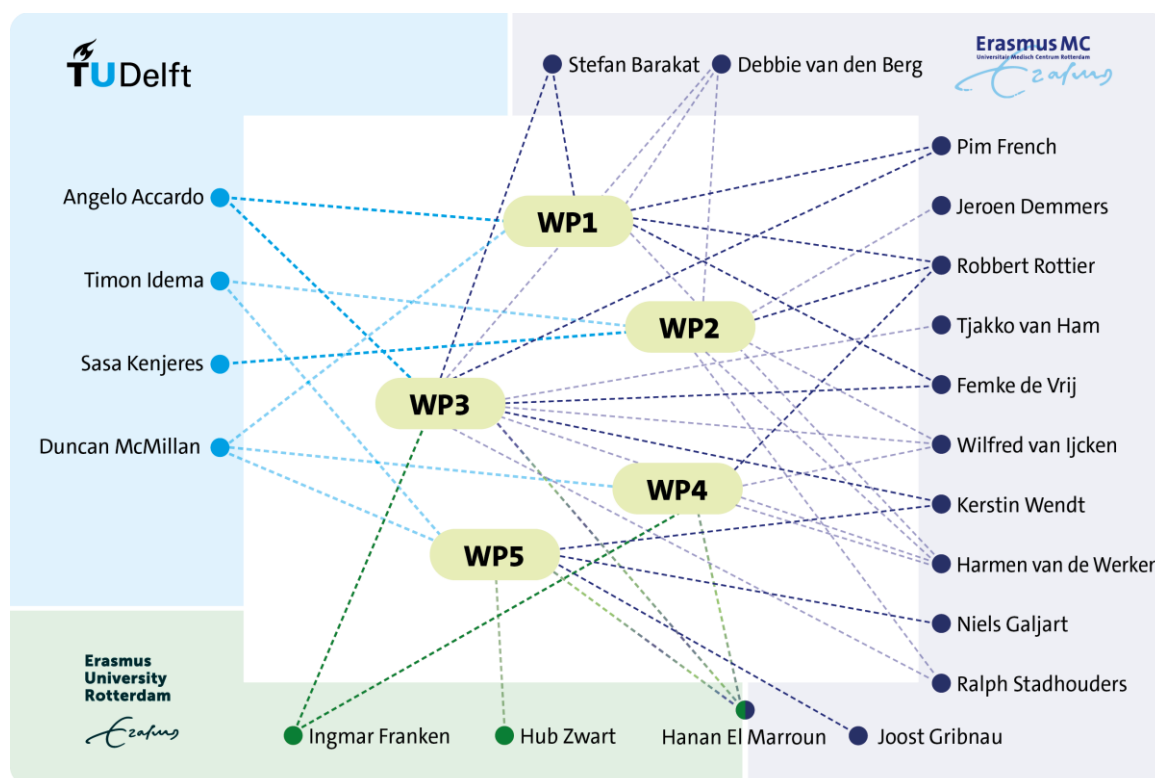


Figure 5: Collaborative network between TU Delft, EUR and EMC

4. Consortium (25%) max 2000 words for 4.a to 4.d

4.a Track record of the consortium members

This consortium is a multidisciplinary team combining expertise from developmental biology, cell biology, genomics, biophysics, micro-engineering, epidemiology, psychology, philosophy and mathematical modelling. Specific details on each team member, see below. All team members have excellent scientific track records, and some of them also translate scientific results into clinical applications/practice. Further, the consortium members have impact on the society in different ways; e.g., knowledge dissemination in teaching, writing books, presenting at or organizing conferences, science communication, role as editorial board members and so on. All consortium members have ample experience in supervision of bachelor, master and PhD students in research projects, as well as teaching and coordinating courses in several programs embedded at the TUD, EMC and EUR. All consortium members have acquired several prestigious national and international grants and thus will be able to attract external funding on the short- and long-term to sustain the flagship.

Short description of all consortium members:

Initials	Publication track record?	Clinical/societal impact	Supervision	Funding
AA	56 publications		>10 BSc/MSc, 5 PhD, 1 WP*	NWO ENW-M-2 and -XS, 3mE Cohesion, Delft Health Initiative
SB	48 publications	Identification novel genetic disorders	>15 BSc/MSc, 6 PhD, 2 WP*	EMBO-LTF, MSCA-IF, Niels Stensen, HFSP-LTF, Erasmus MC, ZonMw Veni, NARSAD YIA, EpilepsieNL Grant, CURE Epilepsy Award, KNAW Early Career Award 2021.
DvdB	18 publication	Contribution to identification novel genetic disorder	>10 BSc/MSc, 2 PhD, 2 NWP*	MRC-CDF, FEBS-LTF, MSCA-IF, EUR
JD	>170 publications			NWO Proteins@work consortium, NWO Roadmap for LRI
IF	>200 publications, editor handbooks		>20 PhD	NWO Veni, Vidi
PF	>100 publications	Clinically relevant prognostic markers. Development of diagnostic gene panels.	2 Clinical Fellows, 1 Post-Doc, 5 PhD, 2 Technicians, 3 interns	KWF, Hersentumorfonds, Westlandse ride, Brain Tumour Charity, Strijd van Salland
NG	>110 publications	Cytoskeleton in CNS and heart disease, and as	>50 BSc/MSc students, 13 PhD	EMBO, KNAW, NWO, ZonMW, KWF, LSBR, ESF, EMC

		target for cytostatic drugs.	students, 7 postdocs	
JG	>350 publications			HFSP, NWO Vidi, Vici, ERC CoG
TvH	40 publications	new tests increasing diagnostic yield for genetic disorders, discovered novel genetic disorders	>20 BSc/MSc, 4 PhD, 1 WP*, 1 NWP*	ZonMW Veni, MSCA-CIG, EUR (x2), Alzheimer stichting fellowship
TI	30 publications	3D bioprinting, open textbooks project	5 PhD, 60 MSc/BSc, 8 iGEM teams	NWO-Rubicon, NWO- Gravity, OCW-SURF
SK	> 290 publications		>180 BSc/MSc, 22 PhD, 2 WP*, 4 NWP*	KNAW, ETH Zurich, MSCA, ZonMW, Hartstichting, MSCA-ETN
HEM	>90 publications	Knowledge used on prenatal substance and medication use and child development (used in clinical guidelines)	>40 BSc/Msc students, 8 PhD students, 3 postdocs	NARSAD YIA, EUR, MSCA- ITN, NWO-Aspasia, ZonMW, multiple charities and foundations
DMM	>45 publications 3 Patents, 3 Pat Pending	Work covered by media Nationally and Internationally (2018-2021). Cofounder 'Macula Scientific Consulting ' and 'Organic Photoglass Ltd'	>38 BSc/MSc, 7 PhD, 14 Industry- linked studentships	NWO, Royal Society NZ and UK, JSPS, FEBS, Stimuleringsfonds
RR	>90 publications	Dissemination of lung development, in vitro culture systems	>15 BSc/MSc, >10 PhD	ZonMw, NWO, Longfonds, SSWO
RS	>50 publications	- Co-founder EMC lung disease patient advisory board - Dissemination of chromosome	8 BSc/MSc, 6 PhD, 2 WP*, 1 NWP*	NWO Veni/Vidi, Longfonds, KWF, Daniel den Hoed Foundation

		conformation capture technology		
FdV	>30 publications in brain disease modeling	- Dissemination/ education stem cell modeling of brain diseases. - Collaborative translational/drug screening studies. - Life and Medical Sciences advisory board Lorentz center.	5BSc/MSc, 4 PhD, 3 Postdoc	ZonMW Create2Solve, ZonMw PSIDER, NWO Gravitation NOCI, Erasmus MC Human Disease Model Award, Horizon2020 FET-PROACT CONNECT
KW	28 publications	SAC member of CdLS World	4 PhD, 2 Postdoc >10 BSc/MSc	NWO BBOL "Genometrack"(coordinator), NWO ALW, E-RARE/ ZonMW "Target CdLS", KWF
HvdW	> 65 publications	Clinical implication of software	3 PhD, 2 Postdocs, > 40 BSc/MSc	DDHS, Health Holland, DUOS.
WvIJ	>250 publications	Cofounder Methylomics B.V.	1 PhD > 20 MSc/BSc	ZonMW, TKI, MRace, NBIC, AIFA
HZ	> 150 papers and 20 books	Academic lead Centre for Society and Genomics (CSG 2004-2013); ethics and society advisor > 10 international projects	20 PhD projects (completed) ; 10 post-doc projects	Coordinator of H2020 project PRINTEGER and Horizon Europe project IANUS; partner in 7 EU projects; CSG; PI in NWO gravitation project BaSyC, 2 NWO/MVI projects
*WP – research assistant, *NWP – trainee research assistant (HBO)				

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The first objective of this flagship is to develop a new generation of innovative 3D models of human tissues. Central to this goal are the use of nano-, micro- and meso-scale structures (e. g. scaffolds, patterning, biosensors) and defined differentiation signals (e. g. growth factors)¹⁷ to create microenvironments that foster tissue self-organization. **AA** has expertise in creating such biomimetic structures that harness mechanobiological responses in neural, cancer and bone cells. **DMM** will integrate biosensors into the 3D-printed micro- and meso-scaffolds manufactured respectively by two-photon polymerisation and stereolithography. In the initial testing and optimisation phase **PF** will contribute his expertise in culturing and imaging of glioblastoma cells on these structures. Next, more complex tissues (i.e., brain and lung) will be grown on our biomimetic structures. **FdV** and **DvdB** use various neuronal differentiation methods (i.e., forebrain organoids,

adherent cortical organoids, 2D cortical networks)^{5,18-22} that can be adapted to bioengineered scaffolds. In addition, **FdV** has ample experience in the generation of pure populations of specific cell types (i.e., astrocytes, oligodendrocytes^{23,24}, microglia) from human iPS cells, allowing sequential addition of specific cell types during guided brain organoid differentiation. **RR** has expertise in lung development, growing airway and alveolar lung organoids, air-liquid interphase culture of lung epithelium and lung on a chip^{3,4,25,26}.

Next, we will focus on dissecting cell-cell communication at different scales (from purified interacting proteins in membranes to cell-cell interactions and cellular ensembles) and in both health and disease states, by constructing biophysical and mathematical models and using, whenever possible, high-dimensional data as input. **FdV** will study synaptic contacts in neuronal networks by whole cell patch clamp recordings and calcium imaging. Physical interactions between cells, for example during neuronal migration or between lung endothelial-epithelial cells, will be imaged in intact, fluorescently labelled mosaic brain organoids (**DvdB**) or lung organoids (**RR**) by lightsheet microscopy (**AA**, **NG**, **DvdB**). **TI** is experienced in using this type of imaging data to build mathematical models of cellular interactions and **SK** will provide his expertise on in silico and mathematical modelling at the level of the entire tissue.

To provide our mathematical model of tissue development with more detailed input on relevant pathways in cell communication, we have gathered a number of experts in spatial mapping and the analysis of nuclear responses to external cues. **RS** will contribute his expertise in single cell mass cytometry and spatial transcriptomics to define active signalling pathways in space and time. **KW** and **RS** have expertise in mapping long-range chromosomal contacts and epigenomic profiling on low cell numbers, enabling the identification of relevant transcriptional regulators and cis-regulatory elements downstream of important signalling cascades. More detailed information on active, functional regulatory elements will be acquired by ChIP-STARR-seq, developed by **SB**. Infrastructure and relevant know-how for acquisition and processing of these different types of -omics data will be provided by **JD** (single cell] proteomics) and **WvIJ** (single cell] genomics). **HvdW** will provide expertise on genomics data analysis and data integration.

At the molecular scale, specific (reconstituted) membrane receptors will be studied on the biomimetic membrane platform developed by **DMM**¹⁶. **KW** has expertise in super resolution microscopy for single molecule imaging and **NG** in studying the interplay between the extracellular environment and the cytoskeleton and how the latter regulates signalling inside cells.

In the second phase of the program (**WP3**), we will investigate disease mechanisms and refine and test predictability of our mathematical models by introducing in our biosystems either pathogenic gene variants linked to structural brain malformations or patient-specific cells derived from bronchopulmonary dysplasia patients²⁷ for which patient (post mortem) data will be provided by **SB**, **TvH** and **RR**. In addition, experimental and in silico data on environmental perturbations (exposure to tobacco, cannabinoids, oxygen, allergens and inflammatory cytokines) will be validated with the expertise of **IF** and **HEM** on neurocognitive and structural brain defects in population-based cohort data and of **RS** on asthma patient parameters. Independent validation will be performed with the expertise of **TvH**²⁸⁻³¹ on modelling genetic and environmental perturbations in zebrafish brain. In particular, mutant/perturbed cells can be studied in in vivo brain development, by fluorescently labeling them, including glial cell types (astrocytes, oligodendrocytes and microglia), as an intermediate in vivo vertebrate brain development model to compare side-by-side with organoid. Similarly, this is also possible for swim bladder development as a proxy for the lung³².

In **WP5**, **HZ** will offer his expertise on philosophy and ethics to develop transferrable skills and broaden the horizon of PhD students engaged in the program, as well as establish a dialogue platform involving researchers, clinicians, patients.

4.c Involvement of partners

In the context of this collaboration, we are exclusively invited by Biopharm HK Ltd (letter of commitment is provided) to examine the binding properties of an exclusively developed drug library to target cannabinoid receptors. The drugs are successive iterations with slight structural changes enabling the opportunity to explore the pharmacological space for exploration against drug targets including cancer, pain, anti-virals etc'. Biopharm HK Ltd is also very interested in the interactions of our various cannabinoids on brain receptors in order to balance the treatment of disease on the effects on the brain. The contribution of Biopharm HK Ltd is not only the supplied drug library, but also further compound development.

4.d Governance, organization and collaboration strategy of the consortium

The flagship will be managed by a board consisting of the three main applicants from each institute (KW, DMM and HEM), completed with four members from the flagship. The latter will be appointed in the board for three years, after which they will be replaced by others. The management board is responsible to maintain the focus and develop the long-term vision of the flagship. Therefore, they will also consult the steering committees of each WP (see below). The management board will meet twice a year.

The management of the WPs will be executed by a steering committee consisting of three chosen members of the WP, complemented with 2 members of other WPs to prevent each WP from becoming an isolated entity. These steering committees will meet monthly to discuss progress and opportunities. Within these meetings, funding opportunities will be discussed, and appropriate action will be initiated if suitable funding opportunities are identified. The steering committee will report to the management board and together decide on future projects. The steering committees will organize an annual scientific meeting of all PI's and students engaged in the BioComm project and will keep track of the project and progress of the PhD students and Postdocs employed by the program

The education and outreach program will be covered by WP5, and will therefore not separately addressed.

The flagship will be advised by an external advisory board consisting of experts in the field that will be installed for 5 years. Members of this committee may be from other Universities, research facilities or companies to ensure a continued connection with the Dutch research environment.

At the start of the flagship project, details of intellectual property and other matters of governance concerning the BioComm flagship project will be detailed in a consortium agreement in collaboration with the Convergence teams.

The members of the flagship will use their individual networks to expand the collaborative efforts initiated by the flagship.

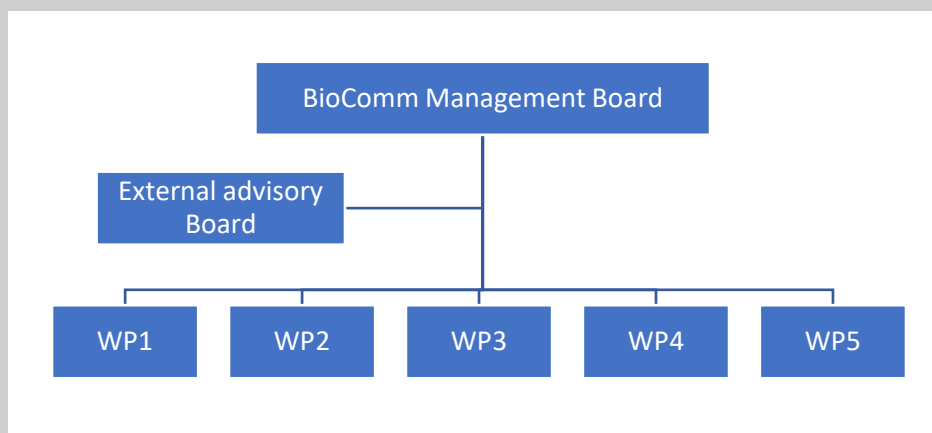


Figure 6: Organogram showing the BioComm governance structure

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Annexes

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

Budget formulier: Convergentie Flagship Call

Flagship:	BioComm
Flagship acronym:	BioComm
Naam Penvoerder:	Kerstin Wendt
Startdatum en looptijd van het project:	1st December 2022- 30st November 2027
Totaal project budget:	€ 2.000.000 by Convergence (€ 400.000 per year) and € 2.878.233 by matching funding

R&D Overview: kostensoorten per type onderzoek (hele project periode)

Loonkosten:										dec22nov23		2023/24		2024/25		2025/26		2026/27		Totaal		Totaal per partner					
Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten			In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind						
Onderzoekorganisaties																											
Erasmus MC	Researchanalist C	1.00	12/1/2022	9/30/2027	Schaal 9	€ 4.479	€ 389.702	€	72.473	€	76.913	€	81.004	€	85.633	€	73.679	€	389.702	€	-	€	389.702.00				
Erasmus MC	Researchanalist C	1.00	12/1/2022	11/30/2026	Schaal 9	€ 4.389	€ 316.030	€	72.473	€	76.913	€	81.004	€	85.633	€	316.023	€	-	€	-	€	316.023.00				
Erasmus MC	OIO	2.00	12/1/2023	11/30/2027	OIO	€ 3.393	€ 488.534	€		€	102.276	€	120.724	€	128.638	€	136.896	€	488.534	€	-	€	488.534.00				
TU Delft	Postdoc	1.00	12/1/2022	11/30/2024	Schaal 10	€ 4.150	€ 149.386	€	72.473	€	76.913	€		€		€		€	149.386	€	-	€	149.386.00				
TU Delft	OIO	0.50	12/1/2022	11/30/2023	OIO	€ 2.766	€ 24.898	€		€	24.898	€		€		€		€	24.898	€	-	€	24.898.00				
TU Delft	OIO	1.00	12/1/2023	11/30/2027	OIO	€ 3.393	€ 244.267	€		€	51.138	€	60.362	€	64.319	€	68.448	€	244.267	€	-	€	244.267.00				
TU Delft								€										€	-	€	-	€	0.00				
Organisatie 4							€ -	€	0									€	-	€	-	€	0.00				
Ondernemingen																											
Onderneming 1						€ -	€ 0											€	-	€	-	€	0.00				
Onderneming 2						€ -	€ 0											€	-	€	-	€	0.00				
Onderneming 3						€ -	€ 0											€	-	€	-	€	0.00				
Onderneming 4						€ -	€ 0											€	-	€	-	€	0.00				
Overige (private) partners																											
Erasmus MC - ZonMw - R.Rottier (awarded grant)	OIO	1.00	12/1/2022	11/30/2024	OIO	€ 3.016	€ 108.571	€	49.795	€	58.776	€		€		€		€	108.571	€	-	€	108.571.00				
Erasmus MC - TAAI Foundation - R.Stadhouders (awarded grant)	Postdoc	0.50	12/1/2022	11/30/2024	Schaal 10	€ 4.150	€ 74.693	€	36.237	€	38.456	€		€		€		€	74.693	€	-	€	74.693.00				
Erasmus MC - VIDI - R.Stadhouders (awarded grant)	Researchanalist C	0.20	12/1/2022	11/30/2024	Schaal 9	€ 4.150	€ 29.877	€	14.495	€	15.383	€		€		€		€	29.878	€	-	€	29.878.00				
Erasmus MC - CSC scholarship - H.v.d.Werken (awarded grant)	OIO	1.00	12/1/2023	11/30/2024	OIO	€ 3.265	€ 58.777	€		€	58.776	€		€		€		€	58.776	€	-	€	58.776.00				
TU Delft - Cohesion grant - A.Accardo (awarded grant)	Postdoc	1.00	12/1/2022	11/30/2023	Schaal 10	€ 4.518	€ 81.315	€	81.315	€		€		€		€		€	81.315	€	-	€	81.315.00				
MSCA Cotand (pending future grant)	OIO	10.00	12/1/2023	11/30/2027	OIO	€ 1.944	€ 1.400.000	€		€	350.000	€	350.000	€	350.000	€	350.000	€	1.400.000	€	-	€	1.400.000.00				
Totaal							€ 3.366.049															€ 3.366.049					
																				Totaal loonkosten		€ 3.366.049					
Kosten van materialen en hulpmiddelen:										dec22nov23		2023/24		2024/25		2025/26		2026/27		Totaal		Totaal per partner					
Partner die de kosten maakt	Materiaal / hulpmiddel						Kosten			In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind						
Onderzoekorganisaties																											
Erasmus MC	Bechtlelee						€ 302.940	€	36.000	€	72.000	€	72.000	€	72.000	€	50.940	€	302.940	€	-	€	302.940.00				
Erasmus MC	Biomica																	€	-	€	-	€	0.00				
Erasmus MC							€ 0											€	-	€	-	€	0.00				
TU Delft																		€	-	€	-	€	0.00				
TU Delft	Dissemination/Webpage						€ 11.000	€	3.000	€	2.000	€	2.000	€	2.000	€	2.000	€	11.000	€	-	€	11.000.00				
Ondernemingen																											
BiPharm	Stargate Library of Compounds						€ 750.000	€	150.000	€	150.000	€	150.000	€	150.000	€	150.000	€	750.000	€	-	€	750.000.00				
Onderneming 2																		€	-	€	-	€	0.00				
Onderneming 3																		€	-	€	-	€	0.00				
Onderneming 4																		€	-	€	-	€	0.00				
Overige (private) partners																											
Erasmus MC - Epilepsie NL - S.Barakat (awarded grant)	Materiaal & consumables Stefan Barakat						€ 100.000	€	50.000	€	50.000	€	50.000	€	50.000	€	50.000	€	100.000	€	-	€	100.000.00				
Erasmus MC - Hersentumorfonds - P.French (awarded grant)	Materiaal & consumables Pm French						€ 40.000	€	20.000	€	20.000	€	20.000	€	20.000	€	20.000	€	40.000	€	-	€	40.000.00				
Erasmus MC - ZonMw - F. de Vrij (awarded grant)	Materiaal & consumables Ferme de Vrij						€ 175.000	€	100.000	€	75.000	€	75.000	€	75.000	€	75.000	€	175.000	€	-	€	175.000.00				
TU Delft - Hartstichting - S.Kenjeres (awarded grant)	Sasa Kenjeres: Hartstichting						€ 35.000	€	35.000	€	35.000	€	35.000	€	35.000	€	35.000	€	35.000	€	-	€	35.000.00				
TU Delft - FF4EuroHPC - S.Kenjeres (awarded grant)	Sasa Kenjeres: FF4EuroHPC						€ 25.000	€	25.000	€	25.000	€	25.000	€	25.000	€	25.000	€	25.000	€	-	€	25.000.00				
Totaal							€ 1.438.940															€ 1.438.940					
																				Totaal kosten materialen en hulpmiddelen				€ 1.438.940			
Kosten van gebruik van machines en apparatuur										dec22nov23		2023/24		2024/25		2025/26		2026/27		Totaal		Totaal per partner					
Partner die de kosten maakt	Machine / apparatuur	Prijs per gebruikseenheid	Gebruiks eenheden				Kosten			In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind						
Onderzoekorganisaties																											
TU Delft	Celris Lumen X						€ 50.000	€	50.000											€ 50.000	€	-	€	50.000.00			
Organisatie 3																				€	-	€	€	0.00			
Organisatie 4																				€	-	€	€	0.00			
Ondernemingen																											
Onderneming 1																				€	-	€	€	0.00			
Onderneming 2																				€	-	€	€	0.00			
Onderneming 3																				€	-	€	€	0.00			
Onderneming 4																				€	-	€	€	0.00			
Overige (private) partners																											
Naam 1																				€	-	€	€	0.00			
Naam 2																				€	-	€	€	0.00			
Naam 3																				€	-	€	€	0.00			
Naam 4																				€	-	€	€	0.00			
Totaal							€ 50.000																€ 50.000				
																				Totaal kosten machines en apparatuur				€ 50.000			
Aan derden verschuldigde kosten										dec22nov23		2023/24		2024/25		2025/26		2026/27		Totaal		Totaal per partner					
Partner die de kosten maakt	Naam derde, omschrijving kosten						Kosten			In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind						
Onderzoekorganisaties																											
Organisatie 1																				€	-	€	€	0.00			
Organisatie 2																				€	-	€	€	0.00			
Organisatie 3																				€	-	€	€	0.00			
Organisatie 4																				€	-	€	€	0.00			
Ondernemingen																											
Onderneming 1																				€	-	€	€	0.00			
Onderneming 2																				€	-	€	€	0.00			
Onderneming 3																				€	-	€	€	0.00			
Onderneming 4																				€	-	€	€	0.00			
Overige (private) partners																											
Naam 1																				€	-	€	€	0.00			
Naam 2																				€	-	€	€	0.00			
Naam 3																				€	-	€	€	0.00			
Naam 4																				€	-	€	€	0.00			
Totaal							€ 0																€ 0				
																				Totaal kosten derden				€ 0			
Publicatie, reis- en verblijfskosten										dec22nov23		2023/24		2024/25		2025/26		2026/27		Totaal		Totaal per partner					
Partner die de kosten maakt	Naam derde, omschrijving kosten						Kosten			In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind						
Onderzoekorganisaties																											
Erasmus MC							€ 12.000		3.000	€	3.000	€	3.000	€	3.000	€	3.000	€	12.000	€	-	€	12.000.00				
EUR																				€	-	€	€	0.00			
TU Delft							€ 11.250	€	3.000	€	3.000	€	3.000	€	1.500	€	1.500	€	11.250	€	-	€	11.250.00				
Organisatie 4																				€	-	€	€	0.00			
Ondernemingen																											
Onderneming 1																				€	-	€	€	0.00			
Onderneming 2																				€	-	€	€	0.00			
Onderneming 3																				€	-	€	€	0.00			
Onderneming 4																				€	-	€	€	0.00			
Overige (private) partners																											
Naam 1																				€	-	€	€	0.00			
Naam 2																				€	-	€	€	0.00			
Naam 3																				€	-	€	€	0.00			
Naam 4																				€	-	€	€	0.00			
Totaal							€ 23.250																€ 23.250				
																				Totaal							



To: Dr Duncan McMillan and Prof Danny Huylebroeck, Head of Cell biology.
Delft University of Technology

Subject: Letter of Support for: **Biosystems modeling of cell-cell communication (BioComm) in Health and Technology Convergence Grant.**

09.03.2022

Dear Dr. McMillan and Prof. Huylebroeck,

With this letter I declare on behalf of BioPharm Limited our interest in and support for your research project proposal **Biosystems modeling of cell-cell communication (BioComm) in Health and Technology Convergence Grant.**

As BioPharm management, we have read and have been fully informed about your application in the **Health and Technology Convergence Grant**. We are very interested in participating in the project.

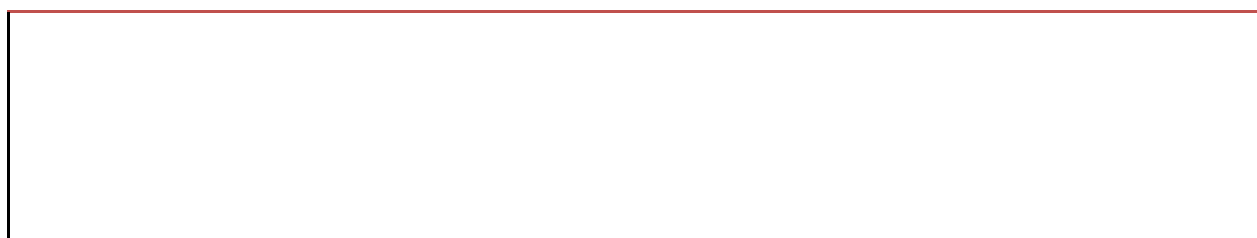
We are a pharmaceutical development company, involved in drug discovery and innovative methods of synthesis and production.

We have previously discovered and patented several hundred compounds in series that we would like to explore against drug targets including cancer, pain, anti-viral etc.

Many compounds with similar structure and pharmacological activity have been found to be therapeutically useful. We are aware that some patient support groups have used a number of our compounds purchased from research chemical market to alleviate pain in certain conditions including stage four cancer suffering, where existing medications had unacceptable side effects profile, and there have been reports in literature of some of our compounds showing favorable activity against some forms of tumor, so we expect that there is likelihood that within our library of compounds there may be a number of hits for potential medical applications. We are also very interested in the interactions of our various cannabinoids on brain receptors in order to balance the treatment of disease on the effects on the brain.

We will contribute to this project. Our contribution will consist of:
An in-kind contribution as itemized in the table below.

Item or activity	Price or rate	Amount
BioPharm Cannabinoid Library of Compounds	EUR750,000	1
Total		



At this stage we are supplying the reference materials produced in purpose built laboratory in New Zealand over a number of years with a development cost of circa EUR750,000

Furthermore, we state that we will actively participate in the User Committee (UC) established and that we will comply with the applicable terms and conditions for participation.

If the project is approved and granted then as a consortium partner we will share responsibility for a prompt completion of a consortium project agreement, based on the template of NWO. As for the arrangements on Intellectual Property and Publication procedure laid down in Annex 2 of the project agreement, I support the decision of the main applicant for choosing option 2: negotiating tailor-made arrangements on IP and Publication procedure.

Yours sincerely,



Matthew Bowden (Government, Media and Outreach Liaison)
BioPharm Limited
21st Floor, CMA Building 64
Connaught Road Central,
Hong Kong
www.biopharm.hk

To whom it may concern

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Fax number +31 10 7044743
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E-mail d.huylebroeck@erasmusmc.nl
Our reference DH/mvg
Date 11 March 2022

LETTER OF SUPPORT
for the
BioComm Flagship PROJECT

Dear

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

I, Prof. Dr. Danny Huylebroeck, in my capacity of head of the Department of Cell Biology at Erasmus University Medical Center (Erasmus MC), which functions as main lead (via Dr. Kerstin Wendt, tenured assoc. prof. (UHD) and group leader/PI at the aforementioned department) of the **BioComm** Flagship project, confirm the strongest possible and very enthusiastic support for this **BioComm** Flagship application and connected research.

Street address
Wytemaweg 80
3015 CE Rotterdam, NL

I am convinced that **BioComm** is a multi-disciplinary, fully integrated, forward looking and visionary, well-assembled multi-partner project that will translate into longer-term medical innovation of human tissue biology, at the molecular and cellular levels, in health and disease. **BioComm** will do so by creating novel tissue-mimetic structures wherein cell-cell communication and its interpretation by the cells (down to single-cell resolution) are studied, with carefully selected focus on brain and lung, but with promise for future expansion towards other organs. Furthermore, it will use fantastic cutting-edge technologies such as in bioengineering (3D bio-printing, biosensors), genome engineering, optogenetics, single-cell omics (including spatial transcriptomics), proteomics (including imaging mass cytometry), imaging (including light sheet microscopy), high-dimensional data analysis and network modelling, and human tissue biology and disease modelling (including organoid and chip-based approaches). **BioComm** can collectively only work out successfully in a Flagship setting and via the described involvement of co-applicants at Erasmus MC, TU Delft and EUR.

Yours sincerely,



Name: Prof. Dr. Danny Huylebroeck
Position: Head of the Department of Cell Biology at Erasmus University Medical Center
Date: March 14, 2022